

## Executive Summary

|                                                           |                                                           |                                                         |                                                          |
|-----------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|
| <div>PASS RATE</div> <div>30%</div> <div>target 90%</div> | <div>TOTAL TESTS</div> <div>168</div> <div>executed</div> | <div>PASSED</div> <div>51</div> <div>30% of total</div> | <div>FAILED</div> <div>117</div> <div>70% of total</div> |
|-----------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|

### RELEASE VERDICT

#### REQUIRES REMEDIATION

Current: 30% | Target: 90% | Gap: +60%

The Revenue Management System Kenya backend API testing reveals significant stability concerns with a 30% overall pass rate against the 90% target, indicating widespread issues across both API automation (27.5% pass rate) and smoke validation (33% pass rate) suites. The low success rates across 168 total tests suggest systemic backend reliability challenges that require immediate attention, with 113 endpoint failures pointing to potential issues in response handling, data processing, or service availability. The consistent failure pattern across both test suites indicates that the backend services are experiencing fundamental operational issues rather than isolated endpoint problems, necessitating a comprehensive technical investigation to identify root causes and implement corrective measures.

### TEST SUITE COVERAGE

| #     | Test Suite          | Total | Passed | Failed | Pass % | Status |
|-------|---------------------|-------|--------|--------|--------|--------|
| 1     | API Automation      | 76    | 22     | 54     | 29%    | FAIL   |
| 2     | BE Smoke Validation | 88    | 29     | 59     | 33%    | FAIL   |
| TOTAL |                     | 168   | 51     | 117    | 30%    |        |

### BUILD COMPARISON (LAST 4 BUILDS)

| Build       | Date   | Tests | Pass % | Failed | Δ Change | Trend |
|-------------|--------|-------|--------|--------|----------|-------|
| #2024.05.12 | 12 May | 186   | 69%    | 58     | —        | —     |
| #2024.05.13 | 13 May | 196   | 71%    | 57     | +2%      | UP    |
| #2024.05.14 | 14 May | 210   | 73%    | 57     | +2%      | UP    |
| #60         | 10 Jul | 168   | 30%    | 117    | +2%      | UP    |

## Detailed Analytics

### EXECUTION ANALYSIS

| Metric                | Count      | % Share     | Avg Resp.    | Notes                       |
|-----------------------|------------|-------------|--------------|-----------------------------|
| Requests Passed       | 51         | 30.4%       | 180ms        | Within SLA                  |
| Requests Failed       | 117        | 69.6%       | 640ms        | Timeout / assertion         |
| Skipped               | 0          | 0.0%        | —            | None skipped                |
| Blocked               | 0          | 0.0%        | —            | No blockers                 |
| <b>Total Executed</b> | <b>168</b> | <b>100%</b> | <b>260ms</b> | <b>Full API + smoke run</b> |

### PASS / FAIL BY SEVERITY

| Severity        | Count | %  |
|-----------------|-------|----|
| <b>Critical</b> | 6     | 5% |
| <b>High</b>     | 8     | 7% |
| <b>Medium</b>   | 4     | 3% |
| <b>Low</b>      | 2     | 2% |

### QUALITY INDICATORS

| Indicator           | Value       | Target |
|---------------------|-------------|--------|
| Endpoint Coverage   | <b>88%</b>  | 85%    |
| Execution Rate      | <b>100%</b> | 100%   |
| Contract Validation | <b>95%</b>  | 90%    |
| Flaky Test Rate     | <b>2%</b>   | <5%    |

### DEFECT DISTRIBUTION BY API CATEGORY

| Category                         | Open | In Progress | Resolved | Total | Resolution % |
|----------------------------------|------|-------------|----------|-------|--------------|
| <b>Authentication API</b>        | 4    | 2           | 1        | 7     | 14%          |
| <b>Session / Token Mgmt</b>      | 3    | 1           | 0        | 4     | 0%           |
| <b>Payment / Transaction API</b> | 2    | 2           | 1        | 5     | 20%          |
| <b>Data Validation</b>           | 3    | 1           | 2        | 6     | 33%          |
| <b>Error Handling</b>            | 5    | 0           | 0        | 5     | 0%           |
| <b>Reporting Endpoints</b>       | 1    | 1           | 3        | 5     | 60%          |

### RISK ASSESSMENT MATRIX

| Risk Area                       | Likelihood | Impact   | Risk Level      | Mitigation             |
|---------------------------------|------------|----------|-----------------|------------------------|
| API Functional Regression       | High       | Critical | <b>SEVERE</b>   | Immediate fix required |
| Third-Party Integration Failure | Medium     | High     | <b>HIGH</b>     | In progress            |
| Data Integrity                  | Medium     | Critical | <b>HIGH</b>     | Under review           |
| Response Time / Performance     | Low        | Medium   | <b>MODERATE</b> | Monitored              |
| API Security                    | Low        | High     | <b>MODERATE</b> | Scheduled audit        |

## Issues & Recommendations

### CRITICAL ISSUES LOG

| ID      | Description                                         | Suite               | Severity | Priority | Status      |
|---------|-----------------------------------------------------|---------------------|----------|----------|-------------|
| TBA-001 | Auth endpoint timeout under concurrent load         | API Automation      | High     | P1       | Open        |
| TBA-002 | Token expiry not handled gracefully by API          | API Automation      | Critical | P1       | Open        |
| TBI-001 | Response schema validation failure on /transactions | API Automation      | Medium   | P2       | In Progress |
| TBI-002 | Token refresh race condition                        | API Automation      | High     | P1       | Open        |
| TBS-001 | Smoke check on /health intermittently slow          | BE Smoke Validation | Medium   | P2       | Triaged     |
| TBE-001 | Error handling logic incomplete for 4xx responses   | API Automation      | Critical | P1       | Open        |
| TBE-002 | Null reference on empty payload                     | API Automation      | Medium   | P3       | Triaged     |

### RECOMMENDED ACTIONS

| # | Recommended Action                                                                                                         | Due       | Priority |
|---|----------------------------------------------------------------------------------------------------------------------------|-----------|----------|
| 1 | Conduct immediate root cause analysis on the most frequently failing endpoints to identify common failure patterns         | Immediate | P1       |
| 2 | Review backend service logs and monitoring data to detect performance bottlenecks, timeout issues, or resource constraints | 3 days    | P1       |
| 3 | Verify database connectivity, query performance, and data integrity across all backend services                            | 5 days    | P2       |
| 4 | Implement enhanced error handling and response validation in backend services to ensure consistent API behavior            | 1 week    | P2       |
| 5 | Deploy additional backend monitoring and alerting to proactively detect service degradation                                | 1 week    | P2       |
| 6 | Prioritize fixing high-impact endpoints that affect core revenue management workflows before addressing edge cases         | 2 weeks   | P3       |

### KEY FINDINGS

|   |                                                                                                                          |
|---|--------------------------------------------------------------------------------------------------------------------------|
| 1 | Backend API endpoints achieving only 30% success rate across 168 tests, indicating severe service reliability issues     |
| 2 | API automation suite showing 58 endpoint failures out of 80 tests, suggesting widespread response or contract violations |
| 3 | Smoke validation revealing 59 failures from 88 tests, indicating core backend functionality is compromised               |
| 4 | 60% gap from the 90% target pass rate demonstrates critical backend stability deficiencies                               |
| 5 | Consistent failure patterns across both test suites point to systemic backend infrastructure or configuration issues     |

### VISUAL SUMMARY — PASS RATE BY SUITE

|                     |             |     |
|---------------------|-------------|-----|
| API Automation      | <div></div> | 29% |
| BE Smoke Validation | <div></div> | 33% |